

SUMMARY OF PRODUCT CHARACTERISTICS

1. NAME OF THE MEDICINAL PRODUCT

TETRACYCLINE MC CAPSULE 250MG

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Each capsule contains 250mg of tetracycline hydrochloride.

For the full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Hard gelatin capsule

Red-yellow hard gelatine capsules size "2"

4. CLINICAL PARTICULARS

4.1. Therapeutic indications

The tetracyclines have a wide spectrum of activity and have been used in the treatment of a large number of infections caused by susceptible organisms. With the emergence of bacterial resistance and the development of other antibacterial agents their use may be restricted increasingly to the treatment of rickettsial infections, including typhus, rocky mountain spotted fever, and Q fever; chlamydial infections, including psittacosis, lymphogranuloma venereum, trachoma and inclusion conjunctivitis and mycoplasmal infections especially those caused by mycoplasma pneumoniae.

Tetracyclines are also used in the treatment of brucellosis, plaque, tularaemia, malabsorption syndromes such as tropical sprue and Whipple disease, urinary-tract infections especially non-specific urethritis and are given in low doses in the long-term treatment of severe acne.

Tetracyclines are used in the management of chronic bronchitis to prevent acute exacerbations due to susceptible organisms.

The tetracyclines are used effectively in the treatment of cholera and reduce fluid requirements.

Relapsing fever responds to treatment with tetracyclines but a Herxheimer-like reactions appears to be a frequent complication; it has been given alone or in conjunction with quinine in the management of malaria especially when resistant to chloroquine.

Tetracyclines have also been given in the treatment of amoebic dysentery, often with an amoebicide and in the treatment of balantidiasis.

In penicillin-sensitive patients tetracyclines have been of value in the treatment of staphylococcal and streptococcal infections where tetracyclines resistance is not a problem, and may also be used in such patients in certain stages of syphilis and in yaws, gonorrhoea, actinomycosis, anthrax and leptospirosis.

Tetracyclines has been used topically in the treatment of chlamydial infections of the eye.

In the treatment of systemic infections the tetracyclines are usually administered by mouth; in severe acute infections they may be given by slow intravenous infusion or by intramuscular injection; parenteral therapy should be substituted by oral administration as soon as practicable.

In children, the effects on teeth should be considered and tetracyclines only used when absolutely essential.

4.2. Posology and method of administration

Method of administration

Capsules are for oral administration. They should be administered one hour before or two hours after, food. Food, antacids containing aluminum, calcium or magnesium and iron-containing preparation may impair absorption of tetracycline. Capsules should be swallowed whole with plenty of water, while sitting or standing.

Posology

The usual adult dosage of tetracycline hydrochloride is 250 or 500 mg every 6 hours preferably one hour before or two hours after meals; up to 3g daily may be given if necessary.

Children have been given 10 to 50mg per kg body-weight daily by mouth.

4.3. Contraindications

Tetracycline is contraindicated in the case of:

- Hypersensitivity to tetracycline group antibiotics or any other of the capsule ingredients.
- Severe renal impairment.
- Children aged less than twelve years old.
- Pregnancy and lactation.

4.4. Special warnings and precautions for use

The use of tetracyclines in children aged under twelve years of age, during the period of tooth development, may cause a permanent grey-yellow-brown discolouration of the teeth. There have also been reports of enamel hypoplasia.

Tetracycline hydrochloride should be used with caution in patients with renal or hepatic impairment, or in conjunction with other drugs having the potential for nephrotoxicity or hepatotoxicity. The risk of kidney failure is enhanced with concurrent use of the anaesthetic agent methoxyflurane.

Tetracyclines have anabolic actions and this may result in an elevation of blood urea nitrogen (BUN).

To avoid excessive systemic accumulation of tetracycline in patients with renal impairment the dosage should be reduced (see Posology and Administration), and in cases of long term therapy it is recommended that serum levels of tetracycline are monitored.

Patients with known hepatic disease should not exceed a daily dosage of 1g (see Posology and Administration).

It is recommended that during long term therapy periodic evaluation of organ systems be performed using appropriate laboratory parameters. This includes the haematopoietic, hepatic and renal systems.

Patients may develop cross sensitivity to tetracyclines.

Bacteria may develop cross resistance to tetracycline antibiotics.

Should signs or symptoms of overgrowth of resistant microorganisms develop, such as candidiasis, staphylococcal enterocolitis, enteritis, glossitis, stomatitis, pruritus ani or vaginitis, the use of tetracycline should be terminated and appropriate measures instituted.

Patients using oral contraceptives should be cautioned that in case of diarrhoea or breakthrough bleeding there is a risk of contraceptive failure.

During administration of tetracycline, patient should avoid direct exposure to natural or artificial sunlight because cases of photosensitivity have been reported.

4.5. Interactions with other medicinal products and other forms of interaction

Drugs:

Aluminium salts: concomitant administration significantly reduces the absorption of tetracycline.

Antacids: these frequently contain aluminium and magnesium salts and concomitant administration significantly reduces the absorption of tetracycline.

Anticoagulants: plasma prothrombin activity is depressed by tetracyclines, therefore lower dosage of concurrent anticoagulants may be required.

Bismuth subcitrate/susallylate: The absorption of tetracycline may be decreased with concurrent administration, it is recommended the doses be staggered and antibiotic efficacy monitored.

Calcium salts: concomitant administration significantly reduces the absorption of tetracycline.

Cholestyramine: Concomitant administration reduces tetracycline absorption therefore tetracycline should be administered one hour before or four to six hours after the cholestyramine.

Cisatracurium/doxacurium: Tetracycline may enhance the neuromuscular blockade effect, dose reduction may be required.

Digoxin: In some patients, coadministration of tetracycline may result in digoxin metabolism inhibition resulting in higher serum levels. Dose adjustment may be required and patients should be monitored for signs of digoxin toxicity.

Iron salts: concomitant administration significantly reduces the absorption of tetracycline.

Isotretinoin: Concomitant administration may result in an increased risk of benign intracranial hypertension.

Magnesium salts: concomitant administration significantly reduces the absorption of tetracycline.

Methoxyflurane: Concomitant administration results in nephrotoxicity, including fatalities. The use of tetracycline in patients undergoing anaesthesia with methoxyflurane should be avoided.

Oral contraceptives: the efficacy of oral contraceptives may be reduced during concomitant tetracycline administration. Patients should be cautioned to use additional non-hormonal contraceptive measures. There may be an increased incidence of breakthrough bleeding (see also Warnings).

Zinc salts: concomitant administration significantly reduces the absorption of tetracycline.

Penicillins: tetracycline should not be used with penicillin.

Food:

Milk: the absorption of tetracycline is significantly reduced by milk or other dairy products.

Laboratory tests:

Glucose in urine: False positive results may be obtained with this test using copper reduction methods. Tests relying on enzymatic glucose oxidase methods do not give false positive results.

4.6. Fertility, pregnancy and lactation

Pregnancy: Animal studies have shown that tetracyclines cross the placental barrier and are found in foetal tissue. There is evidence of toxic effects on the developing foetus, frequently related to skeletal development retardation. In animals treated early in pregnancy there is evidence of embryotoxicity. Tetracycline hydrochloride should not be used in pregnancy unless such use is essential. In such cases, the maximum daily dosage should not exceed 1g. Use of tetracyclines during foetal tooth development, second and third trimesters, may cause a permanent grey-yellow-brown discolouration of the teeth. This can occur during both long term use and following repeated short dose therapy. There are also reports of enamel hypoplasia.

Lactation: Tetracycline is found in breast milk. This may cause permanent tooth discolouration in the developing infant and enamel hypoplasia. If tetracycline is administered during lactation, breast-feeding should be discontinued.

4.7. Effects on ability to drive and use machines

Headache, dizziness, visual disturbances and rarely impaired hearing have been reported with tetracycline administration. Patients should be warned about the possible hazards of driving or operating machinery during treatment.

4.8. Undesirable effects

Infections and infestations:

Frequency not known: overgrowth of resistant organisms (*Candida albicans*, in particular); this may cause glossitis, stomatitis, pseudomembranous colitis (*Clostridium difficile* overgrowth), enterocolitis (caused by resistant staphylococci), rectal and vaginal irritation, inflammatory lesions (with candidal overgrowth) in the anogenital regions (see section 4.4)

Blood and lymphatic system disorders:

Rare: haemolytic anaemia, thrombocytopenia, neutropenia, eosinophilia, agranulocytosis, aplastic anaemia.

Immune system disorders:

Frequency not known: hypersensitivity reactions including Stevens-Johnson syndrome, angioedema, toxic epidermal necrolysis, urticaria, anaphylaxis, anaphylactoid purpura, pericarditis, and exacerbation of systemic lupus erythematosus (see sections 4.3 and 4.8), fixed drug eruptions, exfoliative dermatitis.

Endocrine disorders:

Frequency not known: brown-black microscopic discolouration of thyroid tissue. No abnormalities of thyroid function are known to occur.

Nervous system disorders:

Frequency not known: headache.

Eye disorders:

Frequency not known: visual disturbances, permanent visual loss.

Vascular disorders:

Frequency not known: bulging fontanelles in infants; benign intracranial hypertension in juveniles and adults (see section 4.3). Presenting features were headache, dizziness, tinnitus and visual disturbances including blurring of vision, scotomata and diplopia. Treatment should cease if evidence of raised intracranial pressure develops.

Gastrointestinal disorders:

Rare: dysphagia, oesophagitis and oesophageal ulceration (most of these patients took medication immediately before going to bed)

Frequency not known: gastrointestinal irritations, nausea, abdominal discomfort, vomiting, diarrhoea, anorexia, pancreatitis, permanent tooth discolouration and enamel hypoplasia in children (see sections 4.3, 4.4 and 4.6). Tooth discolouration has also been seen in adults. If gastric irritation occurs, tablets should be taken with food.

Hepatobiliary disorders:

Rare: transient increases in liver function tests, hepatitis, jaundice, hepatic failure.

Frequency not known: hepatotoxicity associated with fatty liver.

Skin and subcutaneous tissue disorders:

Frequency not known: erythematous and maculo-papular rashes, photosensitivity (Patients exposed to direct sunlight or ultraviolet light should be advised to discontinue treatment if any skin reaction occurs), pruritis, bullous dermatoses, skin discolouration.

Musculoskeletal, connective tissue and bone disorders:

Frequency not known: increased muscle weakness in patients with myasthenia gravis (see section 4.4).

Renal and urinary disorders:

Rare: acute renal failure, nephritis.

Frequency not known: raised serum urea, renal dysfunction, especially in patients with pre-existing renal impairment.

4.9. Overdose

Symptoms

There may be nausea and vomiting. Crystalluria and haematuria may occur following very large doses.

Hypersensitivity reactions may occur.

Treatment

There is no specific antidote for tetracycline overdosage. Gastric lavage is recommended. The oral administration of milk or antacids will decrease the absorption of tetracycline. Treatment should be symptomatic and supportive.

5. PHARMACOLOGICAL PROPERTIES

5.1. Pharmacodynamic properties

Pharmacotherapeutic group: Antibacterials for systemic use, Tetracyclines, ATC code: J01AA07.

Mechanism of action: Tetracycline hydrochloride is a broad spectrum bacteriostatic antibiotic. It is taken up into sensitive bacterial cells by an active transport mechanism and binds reversibly to the 30S sub-unit of the bacterial ribosome. This prevents the binding of aminoacyl transfer RNA resulting in inhibition of protein synthesis and thus cell growth.

Microbiology: The following Gram positive organisms are usually sensitive to tetracycline: *Staphylococcus aureus* (some strains), *Staphylococcus sp.* (coagulase negative), *Strep. pneumoniae*, *Strep. pyogenes*, some *Strep. agalactiae* (group B) and some viridans streptococci. The enterococci are resistant. Other sensitive bacteria include strains of *Actinomyces israelii*, *Bacillus anthracis*, *Erysipelothrix rhusiopathiae*, *Listeria monocytogenes*, *Clostridium sp.*, *Propionibacterium acnes*.

The following Gram negative organisms are usually sensitive: *N. meningitidis*, *N. gonorrhoea*, *Acinetobacter sp.*, *Moraxella (Branhamella) catarrhalis*, *Bordetella pertussis*, *Brucella sp.*, *Calymmatobacterium granulomatis*,

Campylobacter sp., *Eikenella corrodens*, *Francisella tularensis*, *Haemophilus influenzae*, *Haemophilus ducreyi* (some strains), *Legionella*, *Pasteurella multocida*, *Streptobacillus moniliformis*, *Aeromonas hydrophila*, *Plesiomonas shigelloides*, *Vibrio cholerae*, *Vibrio parahaemolyticus*. Among the Enterobacteriaceae resistant strains are common, although *Salmonella sp.*, *Shigella sp.* and *Yersinia sp.* are usually susceptible. *Proteus sp.*, *Providencia sp.* and *Pseudomonas aeruginosa* are not susceptible. Some species formerly classed as pseudomonas are susceptible, including *Burkholderia mallei*, *Burkholderia pseudomallei* and *Stenotrophomonas (Xanthomonas) maltophilia*.

Other organisms that are usually sensitive to tetracycline include *Helicobacter pylori*, *Chlamydia sp.*, *Rickettsia sp.*, *Coxiella sp.*, *Borrelia burgdorferi*, *Leptospira sp.*, *Treponema sp.*, *Mycoplasma pneumoniae* and *Ureaplasma urealyticum*.

Organisms with minimum inhibitory concentrations (MIC) up to 4µg/ml are considered sensitive, those with an MIC between 4µg/ml and 12.5µg/ml are considered moderately sensitive.

Resistance to tetracycline is usually plasmid mediated and transferable. With the widespread use of tetracycline resistant strains of the majority of sensitive species have now been reported. The main increases in resistance have occurred among the Enterobacteriaceae, staphylococci and streptococci.

5.2. Pharmacokinetic properties

Tetracycline is incompletely absorbed following oral administration, approximately 75% is absorbed. Absorption is reduced in the presence of food, dairy products and divalent and trivalent metal ions. Administration of 500mg every six hours should produce steady state concentrations of 4µg/ml-5µg/ml. Peak plasma concentrations occur after 1–3 hours, protein binding is 25% to 65%. Elimination half-life is 8–10 hours and is prolonged in renal impairment. There is good distribution into tissues and body fluids with the exception of the cerebrospinal fluid. Bile concentrations exceed blood plasma levels, bronchial secretion levels are about 10% plasma levels. Tetracycline crosses the placental barrier and is excreted in milk.

Approximately 60% of a dose is excreted as unchanged drug in the urine and concentrations of 300µg/ml can be achieved, and a significant route of excretion is via the bile to the faeces. Some enterohepatic reabsorption occurs and complete elimination is slow.

6. PHARMACEUTICAL PARTICULARS

6.1. List of excipients

The capsules also contain pregelatinised starch and magnesium stearate.

The capsule shells contains: gelatin, titanium dioxide (E171), indigo carmine (E132) and erythrosine (E127).

6.2. Incompatibilities

None known

6.3. Shelf life

48 months

6.4. Special precautions for storage

Store below 30°C in the original package, in order to protect from light.

6.5. Nature and contents of container

Capsules are packed in clear PVC/Alu blisters. Packs of 1000 capsules per carton are available. 10 capsules per blister pack, 100 blister packs per carton.

6.6. Special precautions for disposal and other handling

No special requirements.

7. PRODUCT REGISTRATION HOLDER

Komedic Sdn Bhd, 4 Jalan PJS 11/14, Bandar Sunway, 46150 Petaling Jaya, Selangor.

8. MANUFACTURER

Medochemie Ltd. (Far East) - (Oral Facility), 40 VSIP II Street 6, Vietnam-Singapore Industrial Park II, Binh Duong Industry-Service-Urban Complex, Hoa Phu ward, Thu Dau Mot city, Binh Duong province, Vietnam

9. DATE OF FIRST AUTHORISATION/RENEWAL OF AUTHORISATION

17/11/1986

10. DATE OF REVISION OF THE TEXT

October 2022